

Total No. of Questions : 8]

SEAT No. :

P1343

[Total No. of Pages : 2

[4858] - 1087

T.E. (Computer Engineering)

COMPUTER NETWORKS

(2012 Pattern)

Time : 3 Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
- 2) *Figures to the right indicate full marks.*
- 3) *Neat diagrams must be drawn wherever necessary.*
- 4) *Assume suitable data, if necessary.*

- Q1)** a) What is DNS? Explain in brief hierarchical structure of DNS. [6]
- b) Compare link state routing and distance vector routing. [6]
- c) Explain token bucket and leaky bucket algorithm with diagram. [8]

OR

- Q2)** a) Explain POP3 and IMAP4. [6]
- b) Explain socket programming with its primitives. [6]
- c) Draw the packet header format of OSPF and explain in detail. [8]

- Q3)** a) Compare and explain 802.11a, 802.11b and 802.11g wireless standards with respect to [8]
- i) Bandwidth.
- ii) Frequency.
- iii) Signal to noise ratio.
- iv) Range.
- b) Draw and explain WAP architecture. [8]

OR

P.T.O.

- Q4)** a) What is WAE (Wireless Application Environment)? Explain in details. [8]
- b) Write a note on: [8]
- i) Wireless LAN.
 - ii) WML Script.

- Q5)** a) Explain the advantage of DTN over IP network with example. [8]
- b) Draw and explain V1OIP network architecture. [8]

OR

- Q6)** a) What is Vehicular network? What are the challenges and applications of vehicular network? [8]
- b) What is DTN? Explain different layers of DTN. [8]

- Q7)** Write a note on (any 3): [18]
- a) Components of optical network.
 - b) ATM cell format.
 - c) GMPLS.
 - d) SDN.

OR

- Q8)** a) What is Virtualization? List advantages and disadvantages of Virtualization. [6]
- b) Explain how packet switching is different in optical network than Ethernet. [6]
- c) Explain SONET/SDH. [6]

